

VETRIVEL A

AI & Data Science Engineer | Machine Learning | Python | NLP

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PROFESSIONAL SUMMARY

Results-driven AI & Data Science undergraduate (CGPA: 8.0) at Rajalakshmi Institute of Technology with proven experience building end-to-end AI systems in transportation intelligence, NLP, and automation. Skilled in Python, ML algorithms, data visualization, and scalable system design. Actively seeking an internship where I can contribute to real-world AI product development and deepen expertise in applied machine learning and data engineering.

EDUCATION RAJALAKSHMI INSTITUTE OF TECHNOLOGY

Chennai, India | 2024 – 2028

B.Tech – Artificial Intelligence and Data Science

GPA: 8.0 / 10.0

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, SQL

Machine Learning: Regression, Classification, Clustering, Model Training & Evaluation, NLP

Data Analysis: Exploratory Data Analysis (EDA), Data Visualization, Statistical Analysis

Tools & Platforms: Git, GitHub, Tableau, Power BI, Matplotlib, Seaborn, AWS (ML Foundations)

Core Strengths: Problem Solving, Analytical Thinking, Adaptability, Continuous Learning

PROJECTS

AIVA – Artificial Intelligence Interview Validation Assistant

NLP-powered automated interview evaluation platform for candidate assessment and recruiter support.

- Built an NLP pipeline to analyze interview responses, extract semantic intent, and score candidates against role-specific criteria.
- Designed AI models for automated response validation, performance scoring, and candidate benchmarking across multiple interview rounds.
- Developed a web-based recruiter dashboard to streamline hiring workflows and surface data-driven candidate insights.
- Applied text preprocessing, feature engineering, and ML classification to improve assessment accuracy and reduce manual review time.

NAVIS – Intelligent Public Transport Navigation System

AI-driven urban mobility platform with real-time route optimization and explainable recommendations.

- Engineered a multi-criteria route optimization engine using ML models factoring travel time, cost, crowd density, and user preferences.
- Developed crowd prediction and dynamic fare estimation modules with confidence scoring for reliable, context-aware routing.
- Integrated an Explainable AI (XAI) layer to generate human-readable reasoning behind route suggestions, boosting user trust.
- Designed a scalable microservices architecture to support real-world deployment and iterative feature expansion.

CORTEX AI NEXUS – Modular AI Automation Platform

Extensible AI workflow engine for predictive analytics automation and future AI Agent integration.

- Architected a modular AI workflow system with pluggable pipeline components for analytics and process automation tasks.
- Built intelligent decision-support modules capable of ingesting structured and semi-structured data for analytical processing.
- Designed a scalable backend with clean separation of concerns, enabling seamless LLM and AI Agent integration.

CERTIFICATIONS

- Python Programming – IIT Bombay Spoken Tutorial
- AWS Academy Graduate – Machine Learning Foundations
- Artificial Intelligence for All – Infosys Springboard
- Advanced Image Analysis Workshop – ISRO IIRS
- SQL Query Optimization & Data Structures – Simplilearn SkillUp
- AI and Machine Learning Fundamentals – UniAthena

ACHIEVEMENTS & ACTIVITIES

- Delivered 3 end-to-end AI projects spanning NLP, transportation intelligence, and automation within the first year of B.Tech.
 - Earned 6 industry-recognized certifications in AI, cloud computing, and programming from IIT Bombay, AWS, ISRO, and Infosys.
 - Maintaining a CGPA of 8.0 while simultaneously pursuing practical AI engineering experience through independent projects. ▸
- Actively developing skills in intelligent systems design, MLOps practices, and scalable AI product development.